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The Crystal Structure of Polycaproamide: Nylon 6

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INTRODUCTION

The crystal structure of nylon 6, $(—NH(CH_2)_5CO—)_p$, has been investigated in an attempt to explain the differences between its physical properties and those of nylon 66, which has the same proportion of amide groups. A difference of special interest is that the melting point of nylon 6 is 215°C. while that of nylon 66 is 265°C. It was suggested by Hill and Walker[1] that this difference of 50°C. might be due to a lower density of hydrogen bonds in the molecular sheets owing to an unfavorable spatial arrangement of the amide groups (Fig. 1*b*). However, these authors did

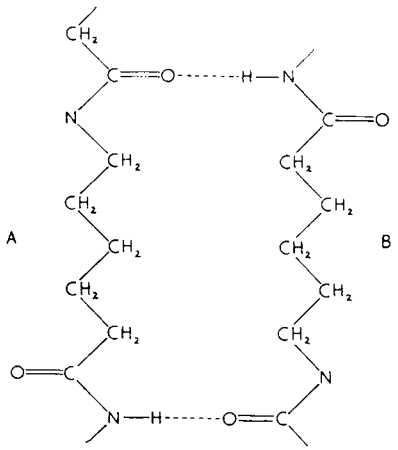


Fig. 1*a*. All hydrogen bonds made perfectly by inverting alternate chains.

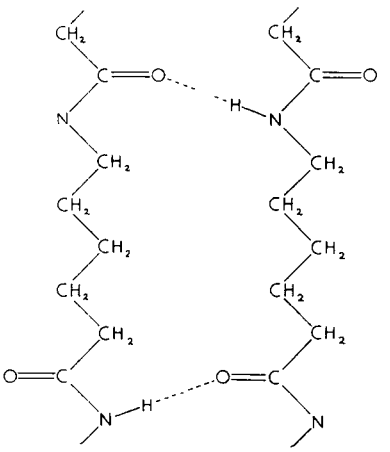


Fig. 1*b*. All hydrogen bonds made imperfectly.

envisage the possibility of making all the hydrogen bonds by inverting alternate molecular chains in the sheet (as in Fig. 1*a*) and such an arrangement was also suggested by Brill[2] as a result of his early work on the crystal structure of nylon 6.

Preliminary x-ray work showed that the polymeric material was fairly crystalline before it was spun; the crystallinity persisted on passage through the extruder and spinneret and was considerably increased on water-conditioning the spun filaments. Undrawn water-conditioned filaments had

no preferred orientation; drawn filaments, on the other hand, obtained in the form of commercial fishing lines or by hand-drawing spun filaments, showed good orientation and crystallinity. The rings on the x-ray photographs localized into short arcs and their breadths diminished slightly showing that the crystallite size had increased. The birefringence increased from 1.8×10^{-4} to 6×10^{-2} for the most highly oriented specimens in good agreement with the previously published refractive indices.[3] The degree of orientation attained is about the same as that in nylon 66 but is less than that in drawn Terylene or stretched polyisobutene. Most samples of polycaproamide contain varying amounts of another crystal form (the β-form) in which the side-by-side packing of hydrogen-bonded sheets is different. The staggered packing scheme along the c-axis is abandoned and all hydrogen-bonded sheets are at the same height in the cell. Details are given of the changes in the x-ray pattern during the conversion of the β- to the normal α-form.

PRELIMINARY X-RAY INVESTIGATION

The investigation of the crystal structure of nylon 6 made by R. Brill led him to conclude that a monoclinic unit cell with dimensions $8.32 \times 9.60 \times 17.2$ and $\beta = 65°$ fitted the x-ray observations best.

We found it difficult to obtain any double orientation similar to that found in pressed polythene[4] and nylon 66[5,6] specimens. Fairly good chain orientation may be easily imposed by drawing spun filaments, but, when some planar orientation is imposed upon these specimens by squashing in a 20-ton press or under a railway train, the original chain orientation is lost. It is likely that the heat generated in squashing is sufficient to destroy the chain orientation and so the resulting strip specimen has only planar orientation. When spun filaments were rolled similar specimens were obtained having only planar orientations. The specimen which showed the highest degree of double orientation was produced by squashing a piece of undrawn monofil in a 30-ton p.s.i. press. Unaccountably some chain orientation had been developed in the direction of the long dimension of the piece of monofil and some planar orientation had also been produced. Unfortunately the x-ray photographs obtained were diffuse, indicating very small crystallite size, but they were improved considerably by annealing the specimens for 15 minutes, at 145°C. and 100% relative humidity. Such photographs proved to be most useful for checking the indexing of the diffraction spots and in elucidating the unit cell but the arc lengths were too long for effective intensity measurements. In spite of the advantages offered by these partially doubly oriented specimens drawn fiber specimens were preferred for intensity measurements as earlier experiments had shown that good orientation could be obtained in conjunction with large crystallite size, giving photographs with short sharp arcs.

Eventually the unit cell proposed by Brill was proved to be approximately correct. The b dimension of the unit cell was redetermined from an oscillation photograph taken with the fiber axis horizontal, which showed the

(0.14.0) reflection; the value 17 24 A. was obtained, in good agreement with the theoretical value 17.30 A. calculated for a planar chain with the accepted bond lengths and bond angles. The dimensions of the basal plane of the reciprocal cell were determined by indexing the spacings of the very sharp equatorial arcs measured on photographs of highly crystalline specimens. The values obtained were $a = 9.56$ A., $c = 8.01$ A., $\beta = 67\frac{1}{2}°$ (*cf.* Brill[2]: $9.6 \times 8.32 \times 17.2$, $\beta = 65°$) and are confirmed by the agreement between calculated and observed spacings for the higher layer lines. There did not appear to be any systematic absences on photographs of fiber or doubly oriented specimens other than $0k0$ for $k = 2n + 1$. There are only equatorial spots for even values of h and k, but this is presumably due to a pseudo-halving in the structure. The space group is therefore $P2_1$, $P2_{1/m}$ or $P2_{1/c}$. The presence of the twofold screw axis is to be expected in view of the geometry of the polymer chain: as the oxygen atom of the carbonyl group is attached to every seventh chain atom, then necessarily successive oxygen atoms will be on opposite sides of a planar zig-zag chain. The form of the chain also makes the existence of mirror planes perpendicular to the chain axis impossible so that the space group is now limited to $P2_1$ or $P2_{1/c}$. As is often the case in polymer crystallography the final choice between these space groups could only be made when the structure was firmly established on the basis of intensity calculations. The crystal structure finally chosen has the lower symmetry space group $P2_1$. The measured density of the piece of monofilament used for the intensity measurements was 1.160 ± 0.001. With the assumption that there are 8 units of the monomer (—$NH(CH_2)_5CO$—) per unit cell, the calculated density of the crystalline regions is 1.23.

INTENSITY MEASUREMENTS

The specimen chosen for the intensity photographs was a piece of monofilament which had been conditioned under tension in steam at 140°C. A series of preliminary heat-conditioning experiments had shown that this specimen had good chain orientation and crystallite size (Fig. 2*a*) and that the amount of unstable crystalline β-structure present (indicated by spots on the second layer-line and elsewhere) had been reduced to a minimum. It was found that the severe heat treatment required to produce large well-ordered crystallites giving fairly streak-free x-ray photographs had a strong adverse effect on the degree of the orientation, sharp x-ray diffraction spots being drawn out into fine arcs along the powder rings. The specimen conditioned in the way described gives the best compromise between disorder and small crystallite size on the one hand and orientation on the other.

A set of photographs on cylindrical films with exposure times of 1 hour, 2 hours, 4 hours, . . .64 hours, were taken with the fiber axis (*b*-axis) mounted vertically. 15° oscillation photographs with the fiber axis mounted horizontally were also taken to record the $0k0$ reflections. The intensities of the diffraction spots were measured by visual comparison with an in-

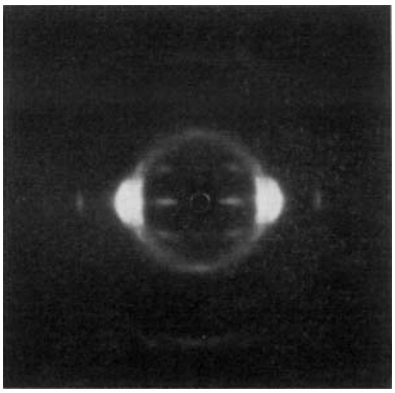


Figure 2*a*.

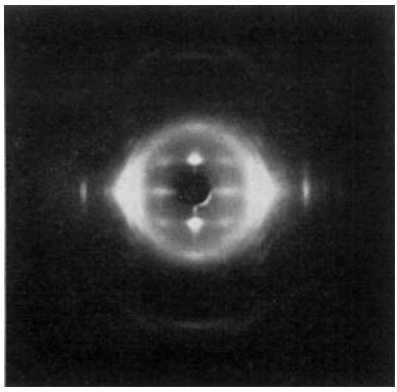


Figure 2*b*.

tensity scale and the usual corrections applied; corrections for spot shape were made by measuring their areas on the strongest photograph, but necessarily the accuracy of the intensity measurements is low owing to the diffuseness and vague limits of a large number of spots.

DETERMINATION OF THE STRUCTURE

The two important pieces of evidence are the length of the *b*-axis and the intensities and positions of the equatorial reflections. The former indicates that the molecular chains are fully extended planar zig-zags, while the equatorial reflections are similar in position and intensity to those of nylon 66,[5] indicating that the structures are alike in their projections on the basal plane. We can at once draw a rough plan of the basal plane (Fig. 3). R. Brill, in a diagram in his paper[2] (Fig. 4), suggests that successive molecules cross the *a*-axis in an alternating fashion but this cannot be so as it would not give the correct sub-unit as deduced from the equatorial absences (shown by broken lines in Fig. 3). For the accurate positioning of the plane of the zig-zag with respect to the (001) plane (*i.e.*, the determination of the angle θ in Fig. 3) we may be guided by the angle determined by Bunn and Garner[5] in the structure of nylon 66 (roughly parallel to the (102) plane of the crystal). The exact determination was made from the evidence provided by the equatorial and other intensities. This method

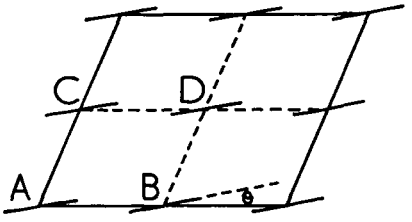


Fig. 3. Basal cell of nylon 6. Pseudocell shown dotted.

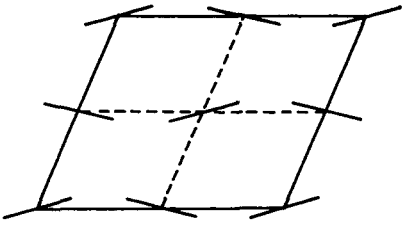


Fig. 4. Basal plane of unit cell of nylon 6 according to Brill. Polymer chains viewed end on.

necessitated the calculation of the equatorial intensities for a large number of orientations of the chain zig-zag with respect to the (001) plane; the labor of these calculations was considerably reduced by rotating an $h0l$ reciprocal lattice net on a one-dimensional Fourier transform graph of the molecule and reading off the structure factors for the various orientations. The value of θ which gives the best agreement between observed and calculated intensities is about 7°, with the chain zig-zag roughly parallel to the (701) set of planes. With this value of θ the hydrogen bond length is 2.81 A., a highly probable figure for an N—H. . .O bond, and there are no other close intermolecular contacts. The hydrogen bond distance, which varies considerably from substance to substance, cannot be used to fix θ, as its calculated magnitude is insensitive to small variations in the angle.

The next point to determine is the arrangement of the chains within the (001) sheets. Chain B (Fig. 3), although identical as viewed along its length, must be different in some way from chain A, otherwise the repeat distance in the a direction would be halved. The most obvious way of fulfilling these two conditions is for chain B to be obtained by inverting chain A; with this arrangement successive chains along the a-axis are oppositely directed. All hydrogen bonds can be made perfectly if the molecules are arranged side by side as in Figure 1a. We may therefore regard the arrangement in the (001) plane as being settled.

The remaining structural feature for investigation is the side-by-side packing of the hydrogen-bonded (001) sheets. Again the results of the nylon 66 structure analysis provide a useful guide to possible sheet displacements in the b-axis direction. The evidence of the $0k0$ intensities alone is sufficient to indicate that a displacement of $^3/_{14}$ b, similar to that found in nylon 66,[5] is highly probable. The considerations outlined above enabled us to draw up four possible trial structures which differed mainly in the relative orientations of successive hydrogen-bonded sheets along the c-axis. All four models had the $^3/_{14}$ b displacement of the carbonyl groups between successive sheets but in some the sheets were reversed left to right and in others top to bottom. Calculations of the expected diffraction intensities of some critical hkl reflection planes showed that only Model 1 gave satisfactory agreement between observed and calculated intensities. In this model the sheet of molecules located at $c = {}^1/_2$ in the cell is an exact duplicate of that at $c = 0$ but is displaced $^3/_{14}$ b in the positive b direction. As Model 1 appeared to be promising, it was used as a basis for a complete calculation of all the hkl diffraction intensities out to the highest sin θ value observed on the strongest photograph, and an R^1-factor (see later) of 27.3% was obtained. Discrepancies still exist, particularly on the 7th layer line, but unfortunately there is no systematic method of refining the atomic positions to try to remove them. Trial and error methods are prohibitive in time and labor in this case, owing to the large number of atoms (thirty-two) in the asymmetric unit.

In the course of the calculations it appeared possible that some improve-

ment in the agreement between observed and calculated intensities might be effected by the introduction of an asymmetric temperature factor to take into account the anisotropy of the thermal vibrations of the long-chain molecules. Such temperature factors have frequently been employed in previous calculations of two-dimensional structure factors for crystal structures in which the atoms are subject to anisotropic thermal vibrations[7,8] but have not hitherto been applied to computations of general *hkl* intensities. However, a simple graphical method was devised for obtaining the necessary temperature factors[9] and a significant reduction in the discrepancies between observed and calculated intensities resulted when they were applied. The R^1-factor decreased to 21.7%.

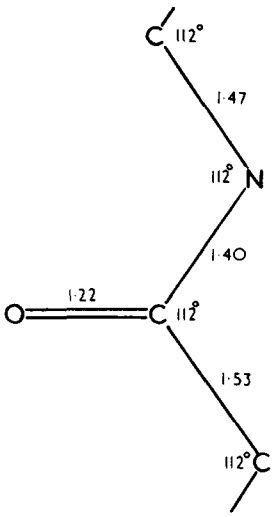


Fig. 5. Chain dimensions used in intensity calculations.

A list of the atomic coordinates used in the final intensity calculations is given in Table II. These coordinates correspond to a molecular model with a planar zig-zag carbon chain, the plane of the zig-zag making an angle of 7° with the (001) plane.

The carbonyl group was assumed to lie in the plane of the chain zig-zag in a direction bisecting the angle between the adjacent C—C and C—N bonds. All chain angles were assumed to be equal to 112° (to simplify the model slightly) and the values used for the other bond lengths and angles are shown in Figure 5.

STRUCTURE OF THE BETA FORM

Certain nylon 6 monofilaments give x-ray photographs which show a very obvious difference from those given by "ordinary" polycaproamide (the α-form). This may be observed by comparing Figure 2*a* with 2*b*: the meridional spot 020 which is barely visible on Figure 2*a* has become the

strongest spot of the photograph in Figure 2*b*. In addition the diffraction spots everywhere on Figure 2*b* and especially on the 7th layer line have become more diffuse and further careful inspection and measurement show that there are many other less obvious differences.

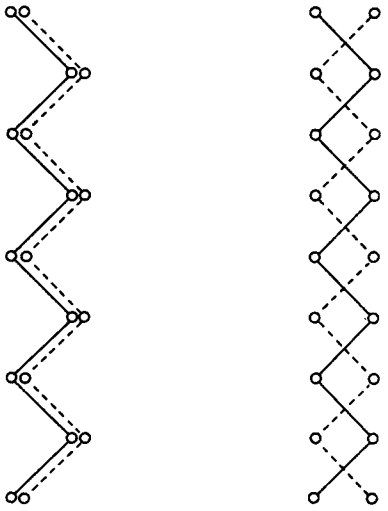
The greatly increased strength of the 020 reflection can only mean that there is a new molecular arrangement in which the carbonyl groups lie at about the same level (equal *y*-coordinates) in the unit cell. This arrangement can be produced in two ways:

(*a*) The $^3/_{14}$ *b* shear between layers is abandoned and successive hydrogen-bonded sheets along the *c*-axis are placed at identical heights in the cell. This gives an arrangement of layers of carbonyl groups, half of them lying at a particular *y*-coordinate and the other half being displaced by only $^1/_{14}$ *b* from this level. In this new arrangement, in which the hydrogen-bonded sheets are retained unchanged, the *c*-axis and unit cell would be halved and only reflections with even *l* values would appear on the x-ray photographs.

(*b*) The $^3/_{14}$ *b* shear between layers is again abandoned but the (020) reflection is given its maximum possible strength by placing all the carbonyl groups at exactly the same level in the cell. This arrangement results in a different sheet structure, rather similar to the earliest models proposed for the α-form (Fig. 1*b*) in which all the molecular chains are similarly directed and all hydrogen bonds are only imperfectly made. Such a structure would produce two experimental effects: (*1*) The x-ray pattern would be simplified as the unit cell would be quartered and only reflections with both *h* and *l* even would appear (*h* and *l* refer to the original cell). (2) The infrared C=O stretching frequency would probably be affected.

From the x-ray photographs it is impossible to decide whether (*a*) or (*b*) is the correct alternative as the β-type specimens all contain a high percentage of the α crystal form. The α structure and both proposals for the β-structure should give identical equatorial intensities but even this is not borne out experimentally as minute examination of Figure 2*b* showed some small differences (unfortunately not visible on the reproduction). Before finally favoring one or other of the proposals we must consider all the available evidence. There is no doubt that the β-form is the unstable one as β-containing specimens could be converted almost completely into the α-form by immersion in boiling water for 5 or 6 hours. Indeed the β-form must be the unstable one for, in both the proposed structures, the similarly charged polar groups are close to each other instead of being approximately uniformly distributed along the *b*-axis as they are in the ordinary α-form and in nylon 66. (The uniform distribution of charged groups is the reason for the shear of $3b/14$ observed in the present structure and in nylon 66.[5]) At first sight a shear of $3^1/_2$ $b/14$ might be considered even more advantageous but this would approach an "uncrossed" structure (Fig. 6*a*) which is less efficient in the packing of CH_2 groups than a "crossed" structure (Fig. 6*b*). These considerations of stability and packing apply with equal force to both the proposed structures for the β-form.

The transition from form (*a*) to the normal α-structure appears to be entirely plausible, merely involving hydrogen-bonded sheets sliding over each other, whereas the transition from form (*b*) would necessitate breaking of hydrogen bonds and either complete inversion of half the molecular chains (extremely unlikely) or rotation through a large angle to form bonded sheets in planes running roughly perpendicular to the original set. This hypothetical transition from the (*b*) form to the stable form by rupture of hydrogen bonds and subsequent rotation of chains cannot be regarded as wholly improbable by reason of the observation that the change only occurs in the presence of water. A specimen containing a considerable amount of the β-structure was maintained dry at a temperature of about 140 °C. for 13 hours and showed no detectable reduction in the (020)



a *b*

"Uncrossed" chains. "Crossed" chains.

Figure 6.

reflection intensity, but gave a considerably improved x-ray photograph in which the crystal reflections had increased in sharpness and intensity at the expense of the amorphous scattering. It is evident that the water acts as a plasticizing agent[6] (nylon 6 swells 12% in water), perhaps by loosening hydrogen bonds in a way similar to its action on cellulosic fibers, and hence allows a readjustment of the crystal structure to occur. Even though the crystal structure alters it is probable that the penetration and plasticization by the water are mainly confined to the amorphous regions as the final changes in the x-ray plane spacings of the crystallites are negligible. To enable the transition between the two crystal forms to occur, it is possible that the small water molecules enter the crystallites, serve as lubricants while the hydrogen-bonded sheets slide over each other, and are then expelled on cooling and drying. No appreciable changes in the crystallite

plane spacings attributable to permanent water penetration have ever been observed in nylon 6 or in nylon 66 or 610.[10,11]

Next the evidence provided by the infrared spectra of films of polycaproamide was considered and here again conflicting results were obtained. In general the spectra could be divided into two classes, A and B, with small differences in the N—H deformation and C=O stretching frequencies and it was natural to suppose that one class would correspond to the α crystal form and the other to the β crystal form. But soon B-type spectra were reported from films cast from α specimens as well as from β specimens and it became apparent that the early observations in which β and α films gave the two different types of spectra were in the nature of misleading accidents. In fact it is now realized that the A type of spectrum occasionally produced from α films is fairly difficult to obtain. At present the existence of the two types of spectra from nylon 6 type polyamides is unexplained and cannot be correlated with the α and β x-ray types. α films which had given B-type spectra (similar to those obtained from β films) were examined by x-rays but in no case did they show the very strong (020) reflection characteristic of the x-ray β type, whereas β films always gave the β-type photographs. It is possible that the film structure is only distantly related to the structure of the original monofil owing to the drastic treatment received in the course of preparation (a typical film was cast in an oven at 140°C. for 1.5 hours after boiling in formic acid for about 3 minutes). This treatment may result in the formation of various intermediate crystallite forms differing in their side-by-side packing of hydrogen-bonded sheets and possibly also in their bonding of adjacent chains. Small randomly occurring layers of these intermediate and disordered forms would explain the streaks appearing on the x-ray fiber photographs. A similar explanation has been suggested for the streaks observed on x-ray photographs of nylon 66 and 610 monofilaments[5] and of some naturally occurring products.[12]

It is obvious that neither of the proposed structures is unanimously supported by the auxiliary evidence, but it seems that proposal (*a*) for the structure is the more reasonable one. Calculations of the intensity changes to be expected on replacing part of the α-structure by the β (*a*) form are in general agreement with the experimentally observed x-ray patterns.

DISCUSSION

The structure of the crystalline portions in polycaproamide fibers is very similar to that of the crystallites in nylon 66 fibers. In both materials the crystallites are made up of hydrogen-bonded sheets and in both the normal method of packing involves a shear of $^3/_{14}$ *b* between sheets (*b* = 17.2 A. for both structures). The appearance of a hydrogen-bonded sheet from each of the two structures is shown in Figures 7*a* and *b*. In nylon 66 (α-form) the shear is progressive (Fig. 8*a*) while in nylon 66 (β-form) and the present structure, nylon 6 (α-form), the shear is staggered leading to a doubling of the axial length in the sheet stacking direction (Fig. 8*b*). The projected cell bases of the two structures are identical except for the

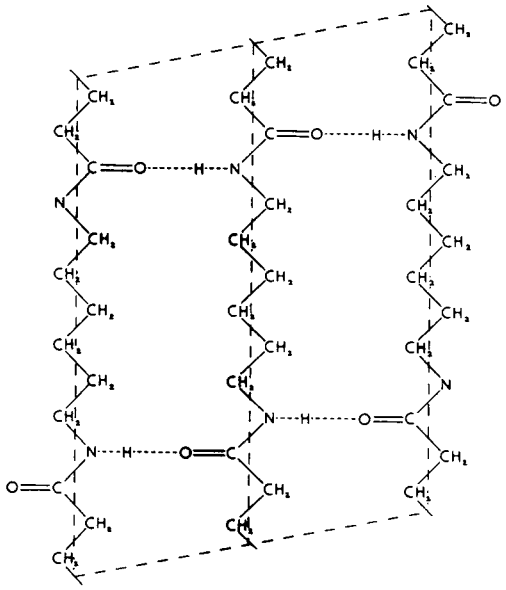


Fig. 7*a*. Hydrogen-bonded sheets in nylon 66. Unit cell face shown dotted.

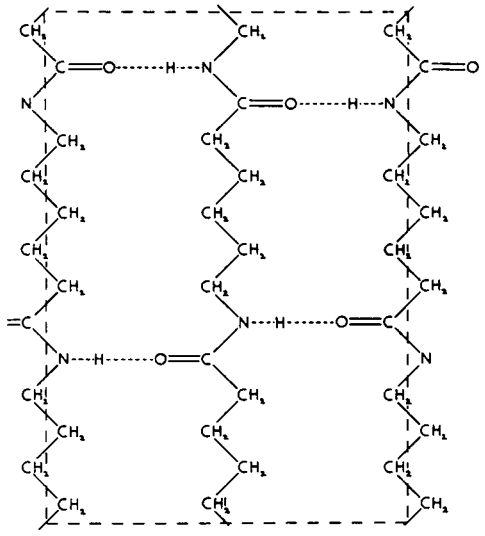


Fig. 7*b*. Hydrogen-bonded sheets in nylon 6. Unit cell face shown dotted.

doubling of both axes which occurs in nylon 6 (α-form). The a-axis is doubled because of the inversion of successive molecular chains to perfect the hydrogen bonding (Fig. 7*b*) and the c-axis is doubled because of the staggered rather than progressive shear. One interesting point of difference between the two structures can be seen by reference to Figures 9*a* and *b*. In nylon 6 the planes of the polymer chains are twisted out of the (001)

plane toward the longer diagonal of the projected cell base, but in nylon 66 they are twisted in the opposite direction toward the shorter diagonal. This difference was so unexpected that great care was taken to see that the angle of tilt suggested by our coordinates really did give the best agreement between observed and calculated intensities. Unfortunately the calculated intensities are rather insensitive to small twists out of the (001) plane (as cos θ is a slowly varying function near $\theta = 0$), but there was an undoubted

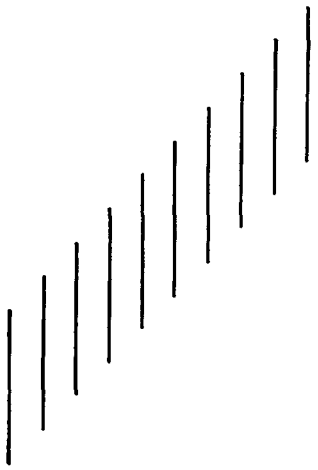


Fig. 8*a*. Progressive shear in nylon 66 α-form.

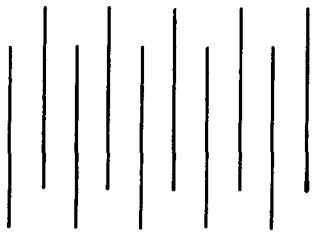


Fig. 8*b*. Staggered shear in nylon 6 α-form and nylon 66 β-form.

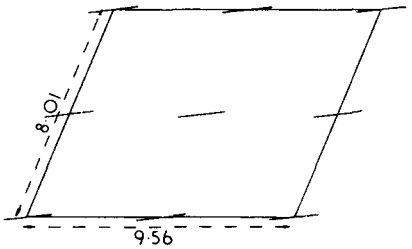


Fig. 9*a*. Nylon 6. Basal plane of unit cell.

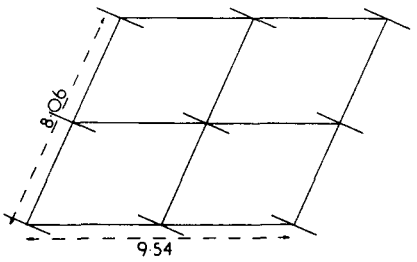


Fig. 9*b*. Projected basal plane of nylon 66. Four unit cells shown.

slight improvement on rotation from the shorter toward the longer diagonal. However, it may be that the experimental evidence is inadequate to decide this point.

If the structure suggested is accepted (and we believe it to be substantially correct) then three further questions are at once raised.

(*1*) Why is the melting point of nylon 6 50°C. lower than that of nylon 66 which has a similar structure and equal density of hydrogen bonds?

(*2*) What is the explanation of the differing side-by-side sheet packing in the two structures, leading to the monoclinic symmetry of nylon 6 α-form

and 66 β-form and the triclinic symmetry of nylon 66 α-form? (See Figs. 8*a* and *b*.)

(*3*) Is there a genuine difference between the chain axis projections of the two structures and if so, why does it exist?

In connection with (*1*) it has been pointed out by Liquori and Mele[13] that if we assume that the heats of melting ΔQ of nylon 6 and nylon 66 are equal, then the observed higher melting point T_m of 66 means that the entropy of melting of 66 ($=\Delta Q/T_m$) is lower than that of 6; and they further suggest that this may be connected with the occurrence, according to Brill,[6] of a transition in 66 crystals at 160°C. thought to be a rotational transition since the packing of the chains becomes hexagonal; in such a rotational transition in the crystal there would be an increase of entropy, so that the entropy of final melting would be reduced. Such a transition was not found by Brill[6] in nylon 6. The two suggestions are not really consistent, because rotation of the molecules would mean breaking hydrogen bonds, so that the heat of melting would be reduced as well as the entropy of melting; the resulting effect on the melting point is impossible to predict. In any case, the approach to an apparent hexagonal packing of chains is not necessarily due to molecular rotation: the expansion normal to the hydrogen-bonded sheets would be expected to be greater than that parallel to the sheets, and this in itself (without any molecular rotation) would lead to pseudo-hexagonal cell dimensions. A similar change occurs in the cell dimensions of nylon 6. As a matter of fact, according to Allen's[14] estimates, both the heat of melting and the entropy of melting of nylon 66 are not less but actually *greater* than those of nylon 6:

	ΔQ	T_m	$\Delta Q/T_m$
Nylon 66.....	22 cal./g.	261°C.	0.0412 cal./°C./g.
Nylon 6.....	19 cal./g.	216°C.	0.0388 cal./°C./g.

It should be remembered that these estimates of ΔQ are rough, and that since they refer to specimens which may contain different proportions of crystalline and amorphous material, they may be misleading. A definite verdict on the question must therefore await more reliable data; but it can be said that there is so far no experimental support for Liquori and Mele's views.

The authors feel strongly that since the phenomenon of melting and the molecular characteristics that control the melting point are not at all well understood, it is unwise to consider particular cases in isolation: it is more profitable to consider these two polyamides first of all in relation to all the other known polyamides and then with other chain compounds. It has already been pointed out by one of us[15] that a diagram of the melting points of the polyamides plotted against the number of amide groups per 100 carbon atoms (Fig. 10) indicates that, apart from the obvious influence of the density of amide groups, the molecular characteristic which appears to control the melting point is the odd or even number of CH_2 groups between the amide groups: those with even numbers melt high, those with

TABLE I

Polyamide type[a]	No. of monomers in crystallographic repeat distance	Molecular symmetry[b] a	b	c	Polar?	Perfection of H bonding
$2m:2n$	1	1	1	$2/m$	No	Yes
$2m:2n+1$	2	$2_1/m$	$1/a$	$2/m$	No	No
$2m+1:2n$	2	$2_1/m$	$1/a$	$2/m$	No	No
$2m+1:2n+1$	1	$1/m$	2	$1/m$	No	No, unless $n = m+1$
$2n$	2	2_1	1	$1/m$	Yes	Yes[c]
$2n+1$	1	1	1	$1/m$	Yes	Yes

[a] n and m are integers. The usual method of describing polyamides is used. The first number is the number of carbon atoms in the diamine and the second the number in the dicarboxylic acid. One number only indicates the number of carbon atoms in the amino acid.

[b] Hermann-Mauguin symmetry symbols are used. The chain-axis direction is labeled a, the direction perpendicular to a and parallel to the planar zig-zag is b. c is perpendicular to a and b.

[c] Means only by inversion of alternate chains in the bonded sheet.

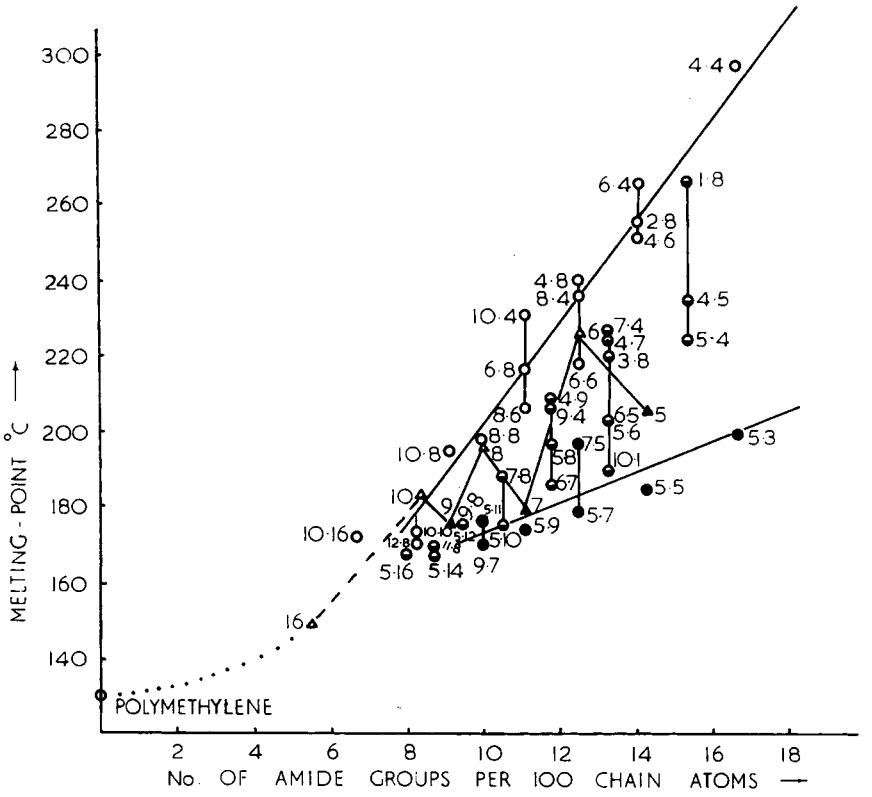


Fig. 10. Polyamides: (O) even number of CH_2 groups and (●) odd numbers of CH_2 groups for diamine-dicarboxylic acid polymers. (◒) Even-odd and odd-even for diamine-dicarboxylic acid polymers. (△) Even number of CH_2 groups and (▲) odd number of CH_2 groups for amino-acid polymers.

odd numbers melt low, while those with mixed odd and even numbers are intermediate. Other molecular characteristics such as symmetry, and crystal characteristics like perfection of hydrogen bonding (*i.e.*, whether CO and HN groups of neighboring molecules are in a straight line or not), show no obvious influence. In amplification of this statement, we give in Table I a list of these characteristics: they show no relation to the distribution of melting points in Figure 10.

The odd-even effect recalls the differences between odd and even members of homologous series which run through the whole of organic chemistry. In short-chain compounds it is always the even numbers which melt higher than the odd ones; and it is perhaps significant that the contrast between even and odd members is much greater in chain molecules having heavy or strongly bound groups at the ends (such as Cl, Br, I, C_6H_5, COOH, $CO\cdot NH_2$) than in the paraffin hydrocarbons; the 50 °C. swing in the dicarboxylic acids is similar in magnitude to the differences between the polyamides. Such facts suggest a common mechanism, possibly connected with the modes of vibration of even or odd chain segments between the anchor points. The effects of the odd or even character on the heats and entropies of melting cannot be predicted theoretically; and empirically, a broad survey of the melting characteristics of molecular crystals in general (both flexible chain molecules and rigid molecules) suggests that at present it is even more

TABLE II

ATOMIC COORDINATES IN POLYCAPROAMIDE EXPRESSED IN FRACTIONS OF THE UNIT CELL EDGES

Molecule A (see Fig. 3 for lettering)				Molecule B			
	x	*y*	*z*		*x*	*y*	*z*
N	0.037	0.285	0.006	N	0.537	0.219	0.006
C	0.040	0.219	0.007	C	0.460	0.285	0.007
O	0.160	0.219	0.007	O	0.340	0.285	0.027
$(CH_2)_1$	0.044	0.146	0.007	$(CH_2)_1$	0.544	0.146	0.007
$(CH_2)_2$	0.040	0.073	0.007	$(CH_2)_2$	0.460	0.073	0.007
$(CH_2)_3$	0.044	0	0.007	$(CH_2)_3$	0.544	0	0.007
$(CH_2)_4$	0.040	0.073	0.007	$(CH_2)_4$	0.460	0.073	0.007
$(CH_2)_5$	0.044	0.146	0.007	$(CH_2)_5$	0.544	0.146	0.007
	Molecule C				Molecule D		
	x	*y*	*z*		*x*	*y*	*z*
N	0.037	0.499	0.506	N	0.537	0.433	0.506
C	0.040	0.433	0.493	C	0.460	0.499	0.493
O	0.160	0.433	0.473	O	0.340	0.499	0.473
$(CH_2)_1$	0.044	0.360	0.507	$(CH_2)_1$	0.544	0.360	0.507
$(CH_2)_2$	0.040	0.287	0.493	$(CH_2)_2$	0.460	0.287	0.493
$(CH_2)_3$	0.044	0.214	0.507	$(CH_2)_3$	0.544	0.214	0.507
$(CH_2)_4$	0.040	0.141	0.493	$(CH_2)_4$	0.460	0.141	0.493
$(CH_2)_5$	0.044	0.068	0.507	$(CH_2)_5$	0.544	0.068	0.507

difficult to correlate heats and entropies of melting with molecular characteristics than to seek correlations directly between melting temperatures and molecular characteristics.

The second point is particularly puzzling as according to our proposals the staggered sheet monoclinic configuration is the stable form in nylon 6 and the unstable form in nylon 66. This situation can only be discussed properly with the aid of models, but from a study of these it appeared that

TABLE III

OBSERVED AND CALCULATED INTENSITIES FOR ALPHA FORM OF NYLON 6[a]

Planes contributing to the reflection	Total calcd. intensity in electrons squared $\times 10^{-3}$	Obs. ntensity $\times 10^{-3}$
200	253	261
002, 202	411	438
$\bar{2}02$, 402	53	44
400	11	7
204	8	13
004, 404	17	27
$\bar{4}02$, 602	2	2
$\bar{2}04$, $\bar{6}00$, 604	4	2
206, 406	1/2	1
006, $\bar{6}02$	1/2	1/2
$\bar{4}04$, 606, 802	3	2
1004, 1002, $\bar{4}06$	3	2
211, 210	5	10
012, $\bar{2}12$, $\bar{2}11$	2	8
120, 021	12	11
$\bar{1}21$, 221, 220	3	4
223, 023, $\bar{3}21$, 222, 323	2	1/2
621, 125, 623, 425	1	1

(continued)

TABLE III (*continued*)

Planes contributing to the reflection	Total calcd. intensity in electrons squared $\times 10^{-3}$	Obs. intensity $\times 10^{-3}$
130, 031, 131	3	2
$\bar{1}31$, 231, 230	3	1
$\bar{2}31$, 331	1	$^1/_2$
$\bar{2}32$, 333, 431, 432	1	$^1/_2$
241, 240, 142, 042, 242	3	1
341, 142, 340, 342	3	1
150, 051, 151	4	3
251, 250, 152, 052, 252	3	1
$\bar{1}52$, 350, 352	1	1
$\bar{3}52$, 154, $\bar{3}54$, $\bar{4}51$, 552, $\bar{5}51$	$1^1/_2$	$1^1/_2$
160, 061	5	$1^1/_2$
361, 162, 360, 362	3	3
$\bar{1}71$, 271	19	58
$\bar{2}71$, 371	40	31
073, $\bar{3}71$, 373, 471	32	8
$\bar{1}73$, 473	1	1
$\bar{4}71$, 571	14	3
573	12	1

TABLE III (*continued*)

Planes contributing to the reflection	Total calcd. intensity in electrons squared $\times 10^{-3}$	Obs. intensity $\times 10^{-3}$
$\bar{5}71$, 375	7	3
773	7	2
$\bar{2}81$, 381, $\bar{1}82$, 380, 382	4	3
020	1	1
040	3	6
060	1/2	1
080	0	0
0100	0	0
0120	0	0
0140	7	4

[a] The factor

$$R' = \frac{\Sigma I_{\text{obs.}} - I_{\text{calcd.}}}{\Sigma I_{\text{obs.}}}$$

for these reflections has the value 0.217. Note that R' is based on intensities and *not* on structure factors. These intensities were calculated on the basis of the atomic coordinates listed in Table I, with the use of a Robertson[16] empirical unitary atomic scattering factor. In addition an asymmetric temperature factor of the form e^{-Bx^2} was superimposed on the spherically symmetrical scattering function to allow for the anisotropic thermal vibrations of the sheet molecules. x is the perpendicular distance in reciprocal angstrom units from the reciprocal lattice point to the plane through the origin normal to the c^*-axis. This means that the c^*-axis, the direction perpendicular to the hydrogen-bonded sheets, was assumed to be the line of greatest thermal vibration. The value chosen for B was 1.9. The contributions of the hydrogen atoms to the scattered intensities were ignored.

the difference in packing might be due to the less regular spacing of the polar groups in nylon 66.

The third point to be considered is the orientation of the planar zig-zag. As the projected reduced cell bases and the lateral dimensions of the chain molecules are very similar in the two nylons it seems that the most probable explanation of the difference reported is lack of adequate experimental evidence. There appears to be no possible reason for a genuine difference in this respect between the structure of nylons 6 and 66.

The authors wish to point out that much of the early work on the structure analysis was done by Miss D. J. Smith. Messrs. R. G. Miller, R. P. Palmer, and P. W. Teare helped with the infrared, optical work, and calculations, respectively.

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Synopsis

The crystal structure of nylon 6 $(—NH(CH_2)_5CO—)_p$ has been determined by interpretation of the x-ray diffraction patterns given by drawn, rolled fibers. The determination was part of a program to investigate the relation between structure and physical properties, in particular melting point. Nylon 6 melts 50°C. lower than its isomer nylon 66 $(—NH(CH_2)_6NH{\cdot}CO(CH_2)_4CO—)_p$; it had been suggested that this was due to deficient hydrogen-bond formation in nylon 6 crystallites. The unit cell contains eight chemical units $(—NH(CH_2)_5CO—)$ and is monoclinic with a = 9.56 A., b = 17.24 A., c = 8.01 A., and $\beta = 67\frac{1}{2}°$. Calculated density = 1.23. Observed density for a drawn monofilament = 1.16. The structure consists of planar chains of CH_2 groups and amide groups tilted 7° from the (001) plane. Alternate chains in this plane are oppositely directed, an arrangement which allows all hydrogen bonds to be made perfectly. The hydrogen-bonded sheets of atoms are packed in an "up-and-down" staggered configuration along the c-axis. Distances between atoms in neighboring molecules are all normal van der Waals contact distances. It appears, from a general survey of polyamide melting points published elsewhere, that the determining factor is the number of CH_2 groups between the amide "anchor points"—polymers with odd numbers of CH_2 groups melt lower than those with even numbers. The present work shows that the odd number of CH_2 groups in this polymer does not lead to deficient hydrogen-bond formation, and that the lower melting point of nylon 6 as compared with nylon 66 must be ascribed to some other cause, possibly connected with the propagation of vibrations along odd-numbered chain segments.

Résumé

La structure cristalline du Nylon 6 $(—NH(CH_2)_5CO—)_p$ a été déterminée par l'interprétation des diagrammes de diffraction aux rayons-X obtenus au départ de fibres étirées. Cette détermination faisait partie d'un programme d'études destinées à mettre en évidence les relations existant entre la structure et les propriétés physiques d'une substance, en particulier leur point de fusion. Le Nylon 6 fond à 50°C plus bas que son isomère le Nylon 66 $(—NH(CH_2)_6NH{\cdot}CO(CH_2)_4CO—)_p$; ceci a été attribué à une déficience dans la formation de liaison hydrogène dans les cristallites de Nylon-6. L'élément périodique contient huit unités $(NH(CH_2)_5CO—)$ et est monoclinique de structure; les valeurs suivantes ont été trouvées pour les divers paramètres a = 9.56 A., b = 17.24 A., c = 8.01 A. et $\beta = 67°1/2$. La densité calculée est égale à 1.23; la valeur trouvée pour un filament isolé étiré est égale à 1.16. La structure répond à des plans de chaînes de groupes CH_2 et de groupes amides tournés de 7° par rapport au plan (001). Alternativement les chaines situées dans ce plan sont orientées en sens inverse l'une per rapport à l'autre; cet arrangement permet la formation de toutes les liaisons-hydrogènes possibles. Les atomes unis par des liaisons-hydrogènes sont réunis en configuration "up-and-down" le long de l'axe c. Les distances entre les atomes des molécules voisines sont les distances normales de contact de Van der Waals. Il semble qu'au départ d'un relevé général des points de fusion des polyamides (publié ailleurs) que la facteur déterminant en est le nombre de groupes CH_2 figurant entre les fonctions amides; il en résulte que les polymères à nombre impair de groupes CH_2 fondent plus bas que ceux à nombre pair de groupes méthylènes. Le présent travail montre qu'un nombre impair de groupes CH_2 dans un polymère n'entraîne pas une déficience dans la formation

de liaison-hydrogèn, et que le point de fusion plus bas du Nylon 6, comparé à celui du Nylon 66 doit être attribué à d'autres causes, peut-être à la propagation possible de vibrations de long de segments de chaînes à nombre impair d'atome le carbone.

Zusammenfassung

Die Kristallstruktur von Nylon 6 $(—NH(CH_2)_5CO—)_p$ wurde durch Auslegung des Röntgenstrahlendiffraktionsdiagrammes einer ausgezogenen, gerollten Faser bestimmt Diese Bestimmung war ein Teil eines Programmes zur Untersuchung des Verhältnisses zwischen Struktur und physikalischen Eigenschaften, insbesondere des Schmelzpunktes. Nylon 6 schmilzt 50 °C. niedriger als sein Isomer Nylon 66 $(—NH(CH_2)_6NH\cdot COCH_2)_4CO—)_p$; es ist vorgeschlagen worden, dass dies durch ungenügende Bildung von Wasserstoffbindungen in Nylon-6-Kristalliten bedingt ist. Die Einheitszelle enthält acht chemische Einheiten $(—NH(CH_2)_5CO—)$ und ist monoklin mit a = 9,56 A., b = 17,24 A., c = 8,01 A. und $\beta = 67^1/_2°$. Berechnete Dichte = 1,23. Beobachtete Dichte für ein ausgezogenes Monofilament = 1,16. Die Struktur besteht aus ebenen Ketten von CH_2-Gruppen und Amidgruppen, die um 7° gegen die (001) Ebene geneigt sind. Alternierende Ketten in dieser Ebene sind entgegengesetzt gerichtet; diese Anordnung erlaubt allen Wasserstoffbindungen, vollständig gebildet zu sein. Die Wasserstoffgebundenen Atom-schichten sind in eine "auf-und-ab" gestapelte Konfiguration an der c-Achse entlang gepackt. Die Distanzen zwischen Atomen in benachbarten Molekülen sind alle normale Van der Waals'sche Kontaktdistanzen. Aus einem allgemeinen Überblick über Schmelzpunkte von Polyamiden, der an anderer Stelle veröffentlicht wurde, geht hervor, dass der bestimmende Faktor die Anzahl an CH_2-Gruppen zwischen den Amid-"Ankerpunkten" ist—Polymere mit einer ungeraden Anzahl an CH_2-Gruppen schmelzen niedriger als die mit einer geraden Anzahl. Die vorliegende Arbeit zeigt, dass die ungerade Anzahl an CH_2-Gruppen in diesem Polymer nicht zu ungenügender Bildung von Wasserstoffbindungen führt, und dass der niedrigere Schmelzpunkt von Nylon 6 im Vergleich mit Nylon 66 einer anderen Ursache zugeschrieben werden muss, die möglicherweise mit der Fortpflanzung von Vibrationen an den Kettensegmenten von ungerader Anzahl in Verbindung steht.

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